

Substrate Non-Algebraic Interfaces Position Document v0.1

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The question

T719 Observable Closure says every BST observable lives in $\overline{Q}(\text{BST primaries})[\pi]$. This is the *algebraic- π -closure* of BST observables.

Question: are there *non-algebraic* interfaces where substrate operates outside $\overline{Q}(\text{primaries})[\pi]$? Where would they appear, and how should we recognize them?

Three candidate non-algebraic interfaces

Interface 1: γ (Euler-Mascheroni)

γ is limit-undecidable — its classification as algebraic vs transcendental is currently open in mathematics itself. Per Casey's " γ as trajectory + catastrophe" framing (memory feedback `gamma_trajectory.md`):

- γ is a *trajectory* not a *number*: $\gamma = \lim_{n \rightarrow \infty} (H_n - \ln n)$
- The limit operation destroys trajectory information (lossy compression)
- γ may sit on the algebraic/transcendental boundary as a fold catastrophe

Substrate-engineering reading: γ is the boundary at which substrate's *limit-taking interface* yields lossy results. The substrate operates via discrete commitment-cycles (per SWPP); γ is what you get when you formally take a limit that the substrate doesn't natively compute.

BST observables involving γ : thermodynamic constants, certain spectral asymptotics, Selberg zeta normalizations. These would be partial-algebraic observables — algebraic up to a γ correction.

Interface 2: $\zeta(2k+1)$ (odd zeta values)

$\zeta(2) = \pi^2/6$ is algebraic in π . $\zeta(4), \zeta(6), \zeta(8), \dots$ all algebraic in π .

But $\zeta(3), \zeta(5), \zeta(7), \dots$ are conjecturally transcendental and algebraically independent from π . Apéry proved $\zeta(3)$ irrational; nothing else known.

Substrate-engineering reading: even-zeta values are *substrate-internal* (algebraic in π). Odd-zeta values appear when substrate interfaces with non-substrate temporal structure — e.g., experimental observables that don't reduce to substrate's commitment-cycle.

BST observables involving $\zeta(2k+1)$: certain QFT loop corrections, statistical ensemble averages. T1930 confirms $\zeta(g)$ -something $\approx$ specific particle property — but $g=7$ puts us in odd-zeta land.

Interface 3: Continued-fraction-algebraic numbers (non-quadratic irrationals)

Quadratic irrationals ($\sqrt{d}$) have eventually-periodic continued fractions. Higher-degree algebraic numbers do not.

Substrate-engineering reading: substrate's K75 Stark anchor uses class-number-1 imaginary-quadratic fields $\mathbb{Q}(\sqrt{-d})$ for $d \in \{3, 7, 11, \dots\}$. These are *substrate-canonical*. Higher-degree algebraic interfaces (cubic, quartic) would be rare-anchor observables that need extra substrate scaffolding.

Test: do BST predictions ever require degree-3+ algebraic numbers? If not, this is structural evidence for substrate's quadratic-irrational preference.

The three-interface taxonomy

Interface	Algebraic?	BST role	Test
1. γ	undecidable	thermodynamic/asymptotic boundary	look for γ in BST D-tier
2. $\zeta(2k+1)$	conjecturally transcendental	temporal-experimental boundary	look for $\zeta(\text{odd})$ in BST D-tier
3. $\text{degree} \geq 3$ algebraic	algebraic but not quadratic	rare-anchor scaffolding	look for cubic+ numbers in BST D-tier

Substrate Closure check (per K137 / Substrate Closure Principle)

Substrate Closure says substrate is operationally closed; no external simulator. The three non-algebraic interfaces above are *not* external simulator referents — they are internal boundaries where substrate’s native algebraic computation produces lossy results.

This is consistent with Substrate Closure: the substrate doesn’t reach for external infinity; it produces *partial* outputs at its boundary interfaces, and we should expect observable correction terms in γ , $\zeta(\text{odd})$, or higher-degree algebraic numbers to mark those boundaries.

Cross-Cartan extension (per K141 / 3306× empirical)

If we test the three-interface taxonomy on alt-HSDs (D_I, D_II, D_III): - D_I should have *different* non-algebraic interface set (different Bergman kernel structure) - D_IV⁵ specifically maps γ to thermodynamic boundary, $\zeta(2k+1)$ to temporal, etc.

If alt-HSDs map non-algebraic interfaces *differently* and BST observables match the D_IV⁵ specific mapping (not alt-HSD), that’s additional substrate-uniqueness evidence.

Pending verification

1. Catalog scan: which BST D-tier observables contain γ ? $\zeta(2k+1)$? degree-3+ algebraic numbers?
2. Lyra theorem: T2456 candidate — “BST observables interface non-algebraic structure only at three boundary types”
3. Strong-Uniqueness criterion: if alt-HSDs fail to map the three-interface taxonomy correctly, this becomes C18 (RIGOROUSLY CLOSED candidate)

Status

Substrate Non-Algebraic Interfaces Position v0.1 filed Friday 08:44 EDT (FAST CADENCE). Closes Wednesday substrate review owed #3. Six-interface cross-cartography map remains owed (Task #233) — that’s the broader synthesis.

— Keeper, position v0.1 Friday 08:44 EDT (FAST CADENCE)